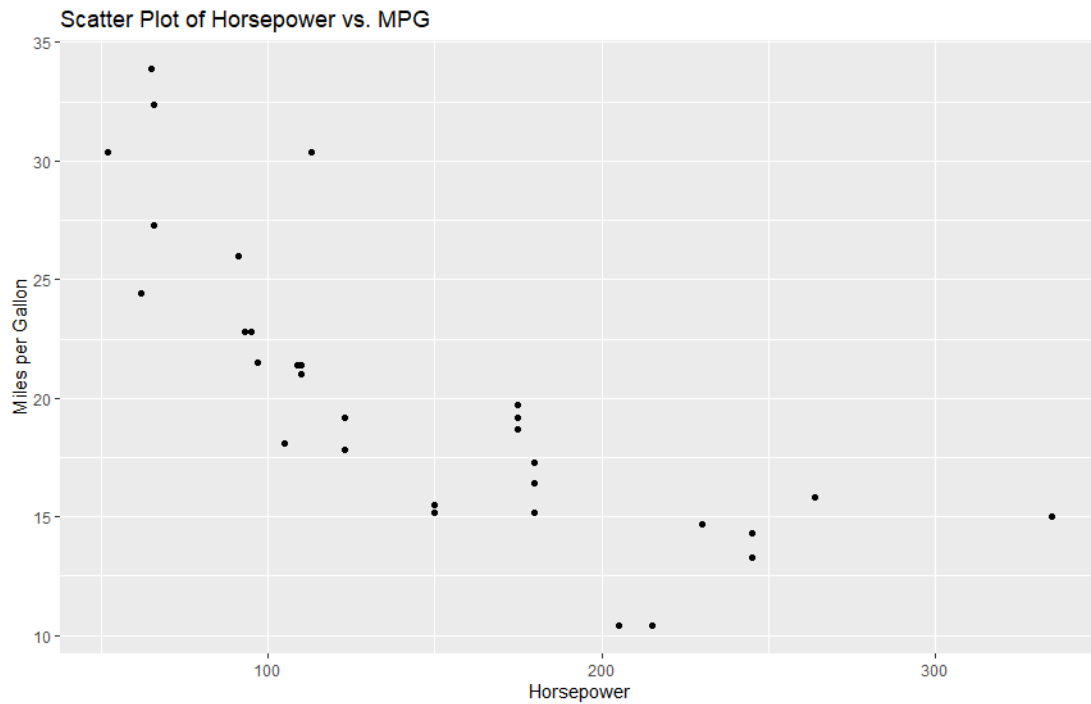


Week 7 : Data VisualisationFile

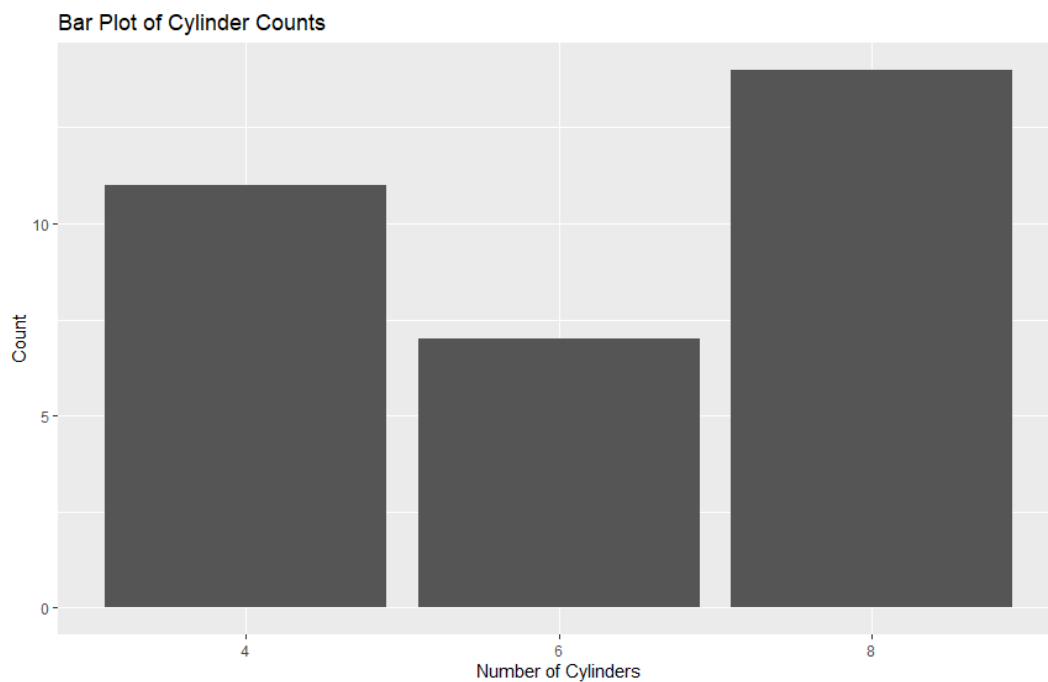
Scatter Plot

```
ggplot(mtcars, aes(x = hp, y = mpg)) +  
  geom_point() +  
  labs(title = "Scatter Plot of Horsepower vs. MPG",  
        x = "Horsepower",  
        y = "Miles per Gallon")
```



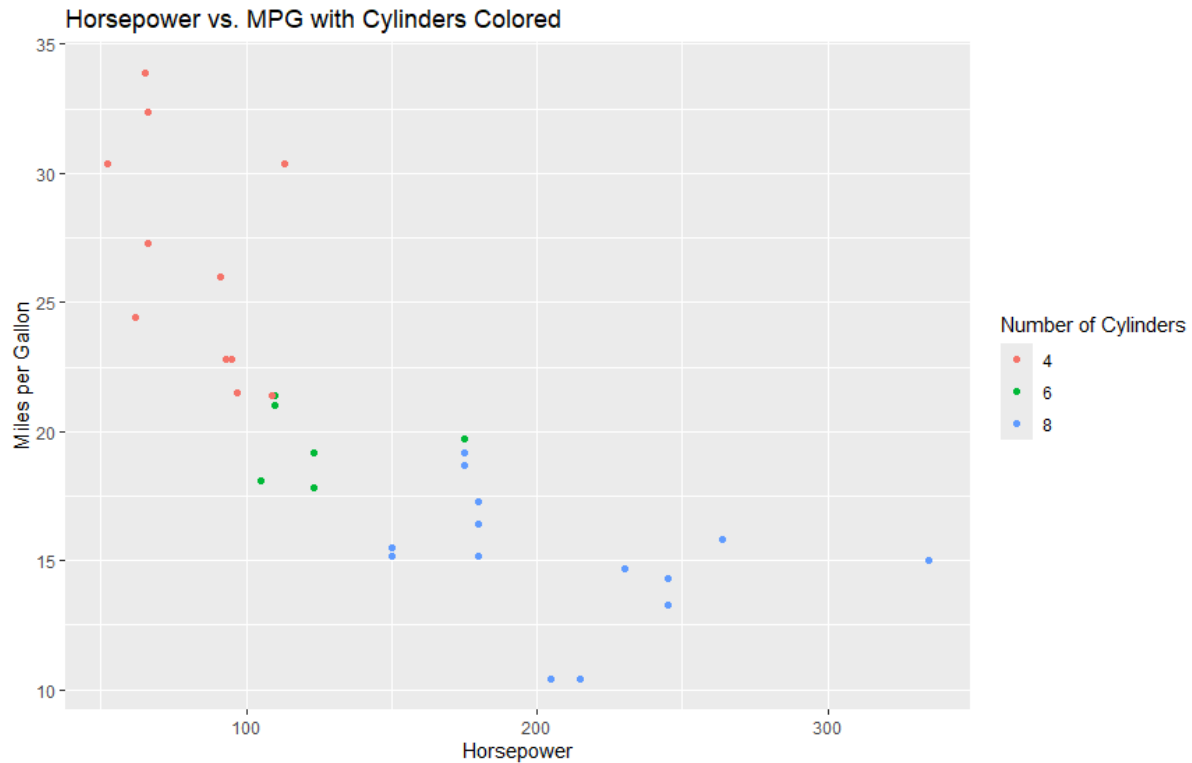
Bar Plot

```
ggplot(mtcars, aes(x = factor(cyl))) +  
  geom_bar() +  
  labs(title = "Bar Plot of Cylinder Counts",  
        x = "Number of Cylinders",  
        y = "Count")
```



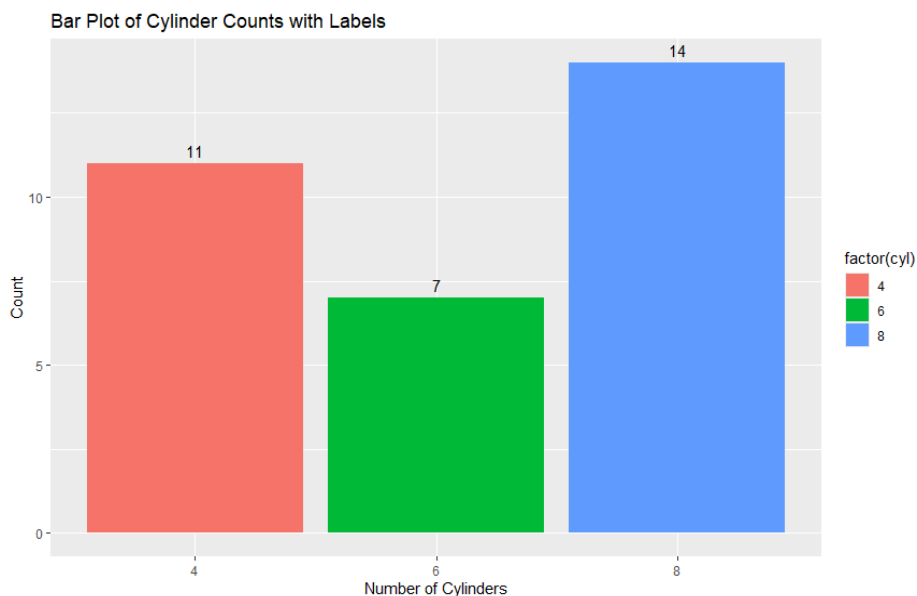
Scatter Plot with Color

```
ggplot(mtcars, aes(x = hp, y = mpg, color = factor(cyl))) +  
  geom_point() +  
  labs(title = "Horsepower vs. MPG with Cylinders Colored",  
        x = "Horsepower",  
        y = "Miles per Gallon",  
        color = "Number of Cylinders")
```



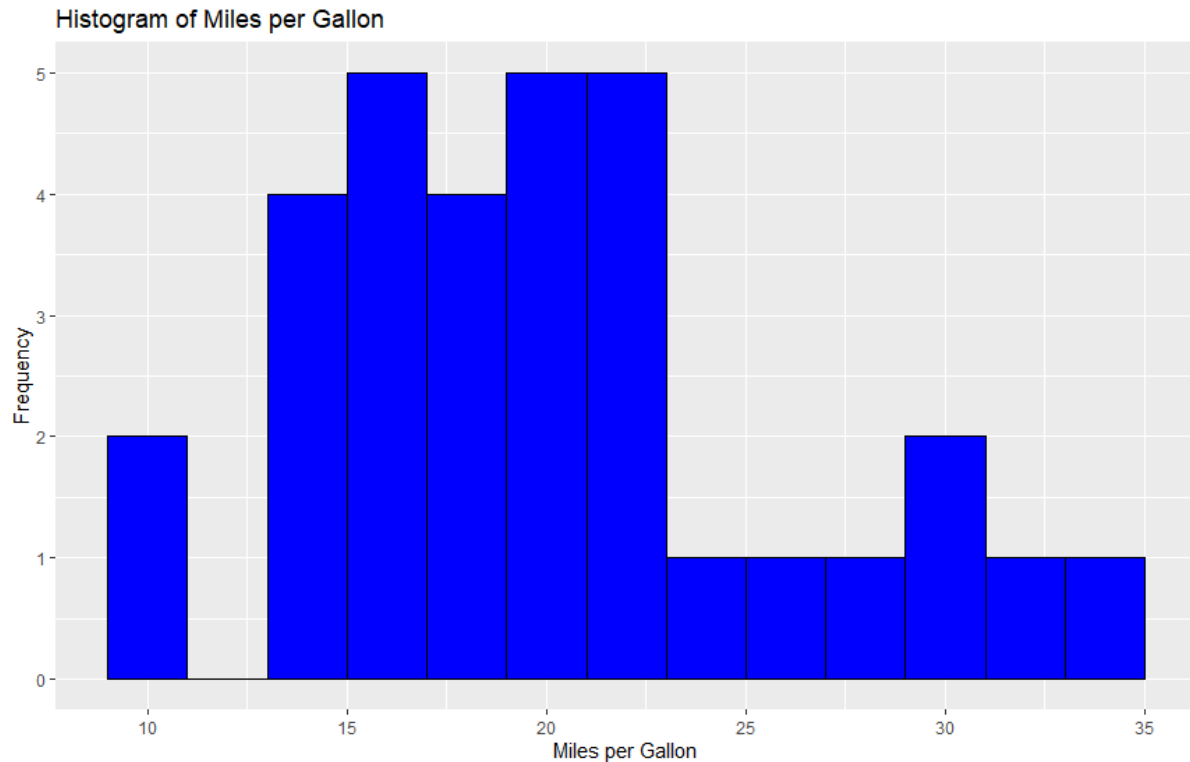
Bar Plot with Labels

```
ggplot(mtcars, aes(x = factor(cyl), fill = factor(cyl))) +  
  geom_bar() +  
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5) +  
  labs(title = "Bar Plot of Cylinder Counts with Labels",  
        x = "Number of Cylinders",  
        y = "Count")
```



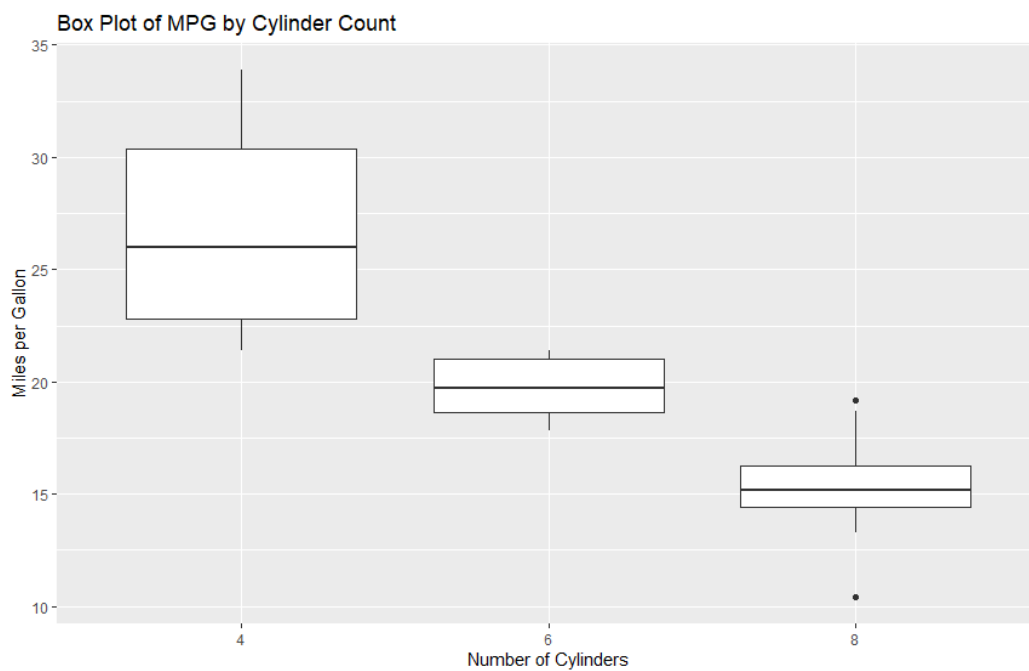
Histogram of Miles per Gallon

```
ggplot(mtcars, aes(x = mpg)) +  
  geom_histogram(binwidth = 2, fill = "blue", color = "black") +  
  labs(title = "Histogram of Miles per Gallon",  
        x = "Miles per Gallon",  
        y = "Frequency")
```



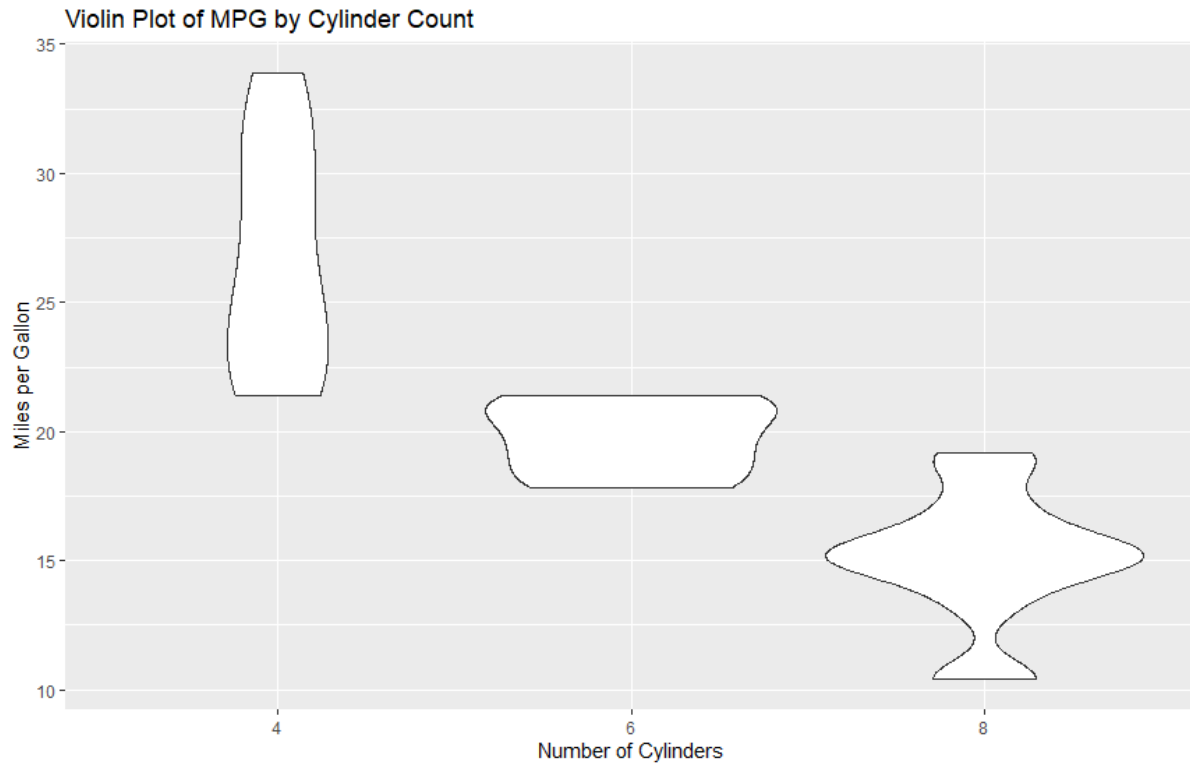
Box Plot of MPG by Cylinder Count

```
ggplot(mtcars, aes(x = factor(cyl), y = mpg)) +  
  geom_boxplot() +  
  labs(title = "Box Plot of MPG by Cylinder Count",  
        x = "Number of Cylinders",  
        y = "Miles per Gallon")
```



Violin Plot

```
ggplot(mtcars, aes(x = factor(cyl), y = mpg)) +  
  geom_violin() +  
  labs(title = "Violin Plot of MPG by Cylinder Count",  
        x = "Number of Cylinders",  
        y = "Miles per Gallon")
```



Line Plot

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_line() +  
  labs(title = "Line Plot of Weight vs MPG",  
        x = "Weight",  
        y = "Miles per Gallon")
```



Density Plot

```
ggplot(mtcars, aes(x = mpg)) +  
  geom_density(fill = "lightblue") +  
  labs(title = "Density Plot of Miles per Gallon",  
        x = "Miles per Gallon",  
        y = "Density")
```

Density Plot of Miles per Gallon

